

OneTapp Analytics Assistant

Dataset link: <https://www.kaggle.com/datasets/beatafaron/fmcg-daily-sales-data-to-2022-2024>

Report purpose	Scope
Document the current MVP and define the work required for a governed production release.	Natural-language FMCG analytics across sales, promotion, inventory, SKU, category, channel, and trend questions.

1. Executive Summary

OneTapp is an AI-assisted FMCG analytics application that lets business users ask natural-language questions and receive governed SQL-backed answers. The MVP demonstrates a complete analytical path: question intake, intent planning, semantic retrieval, SQL generation, validation, structured explanation, visualization and downloadable data.

The MVP is appropriate for demonstrations, assessment and controlled internal prototyping. A production release should retain the same core flow but replace local data access and heuristic safeguards with governed warehouse access, identity-aware authorization, stronger evaluation, observability and operational controls.

2. Current MVP: What Has Been Developed

Capability	MVP implementation	Business value
Conversational analytics	Streamlit chat interface with an analysis catalog, recent-query shortcuts, period selection and CSV export.	Business users can explore data without writing SQL.
Question planning	Planner classifies descriptive, comparative, diagnostic and ambiguous requests and requests clarification when needed.	Reduces accidental interpretation of vague business questions.
Semantic grounding	Semantic metadata describes tables, columns and certified business metrics; retriever selects relevant context.	Connects language to business meaning rather than raw schema alone.
Text-to-SQL	LLM-backed generation with deterministic fallback templates for revenue, promotion, stock, SKU, category, channel, brand and trends.	Keeps core demo flows usable even when an LLM is unavailable.

Safety and execution	SQLGlot parser validation, allowed-table controls, schema checks, metric checks, repair attempts and DuckDB execution.	Improves safety, traceability and correctness before results are shown.
Business response	Narrative summary, key metric, comparison context, interactive chart controls, data table and SQL/agent trace.	Makes analytical output understandable and auditable.
Quality controls	Gold evaluation cases and automated tests cover core flows, validator behavior, period filtering and analysis-catalog queries.	Provides regression protection during iteration.

3. MVP Architecture

The application is organized as a stateful, multi-stage pipeline. Each stage has a specific responsibility, which makes the flow easier to test and explain than a single unrestricted LLM call.

- User interface: Streamlit accepts questions, offers analysis templates, applies an optional year filter, caches repeated work and presents charts and tables.
- Planner: extracts entities and selects an intent archetype; ambiguous questions trigger a clarification response.
- Schema retriever: loads semantic metadata and supplies relevant table, column and business-metric context.
- SQL generator: uses an LLM when configured and controlled fallback templates otherwise.
- Validator and repair loop: parses SQL, checks its structure and references, validates approved metrics and requests bounded repair when validation fails.
- Execution and synthesis: executes approved SQL against DuckDB and converts the result into a concise business response.

4. MVP Scope and Known Limitations

Area	Current MVP position	Limitation
Data platform	Local CSV files loaded into DuckDB.	Appropriate for a prototype; not a managed enterprise data environment.
Access control	No user identity or authorization model.	No row-level security, column masking, or audit trail by user.
Semantic retrieval	Metadata-driven retrieval with business definitions and keyword/synonym matching.	Not yet an embedding-backed vector retrieval service with formal ownership and change control.
SQL coverage	Strong templates for common FMCG questions plus LLM generation.	Fallback paths remain heuristic and dataset-specific for unusual phrasing or complex joins.

Validation	Parser, schema, and selected metric checks; bounded repair loop.	Cannot guarantee every SQL query is semantically correct without broader test coverage and approved query patterns.
Operations	Local caching and automated tests.	No production telemetry, service-level objectives, alerting, or cost controls.

5. Production-Grade Target Architecture

A production release should preserve the user-facing experience while changing where trust is enforced. The data platform, not the LLM, must be the authority for access, governance and aggregation.

Layer	Production-grade capability	Outcome
Identity and governance	OAuth2 or SSO session identity; warehouse RBAC; row-level security and dynamic column masking.	Every answer is limited to the user's approved data scope.
Data platform	Snowflake or Databricks/Unity Catalog; curated tables, dbt transformations, materialized business metrics.	Reliable, scalable and governed analytical execution.
Semantic layer	Versioned metric catalog, documented joins, metric ownership, embeddings/vector retrieval and approved synonyms.	Consistent business definitions and better context retrieval.
Query safety	AST policy enforcement, live catalog validation, cost/scan/time limits, query tagging, parameterized filters and allowlisted joins.	Prevents unsafe, expensive, or semantically unsupported queries.
LLM operations	Model routing, prompt/version management, response caching, rate limits and fallback policy.	Predictable cost, latency and quality.
Observability	Structured logs, traces, audit records, user feedback, quality dashboards and alerting.	Ability to detect failures, misuse, drift and declining answer quality.
Human controls	Approval workflow for high-stakes metrics and escalation to analysts when confidence is low.	Appropriate safeguards for executive or financial decisions.

6. Production Delivery Roadmap

Phase	Priority deliverables	Exit criteria
1. Stabilize MVP	Resolve remaining UI text/encoding issues; consolidate package/import layout; expand gold evaluation set; add LLM-mocked integration tests.	All supported catalog questions are tested and no known user-facing errors remain.
2. Govern data access	Move to a governed warehouse; implement SSO, RBAC, RLS, masking, query tagging and audit logging.	Data access is identity-aware and policy-enforced by the platform.

3. Strengthen accuracy	Version semantic metrics and joins; build an embedding-backed retriever; add semantic-result tests and confidence thresholds.	Measured accuracy meets agreed business thresholds on a representative benchmark.
4. Operate at scale	Introduce service APIs, asynchronous jobs where needed, caching, monitoring, alerting, cost controls and incident runbooks.	Defined reliability, latency and cost targets are consistently met.
5. Expand adoption	Role-specific dashboards, feedback loops, analyst escalation and additional data domains.	Usage and feedback demonstrate repeatable value across business teams.

7. Recommended Success Measures

- Analytical quality: execution success rate, semantic accuracy on gold questions and clarification success rate.
- Business value: reduction in analyst turnaround time, self-service query adoption and repeat use by business users.
- Reliability: p50/p95 response latency, query failure rate, data freshness, and service availability.
- Governance: percentage of queries executed under identity-aware policy, audit coverage and blocked unsafe-query rate.
- Cost: model cost per successful answer, warehouse cost per query, cache hit rate and high-cost query frequency.

8. Conclusion

OneTapp Analytics Assistant demonstrates the essential mechanics of a trustworthy conversational assistant: grounding in business metadata, structured planning, safe SQL execution, repair, explainability, and analyst-friendly output. The production path is not a rewrite; it is a controlled evolution from local prototype components to governed enterprise services, with security, measurement and semantic quality made explicit at every stage.